

14

Firms in Competitive Markets

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What is a Competitive Market? Part 1

- Competitive market
 - Perfectly competitive market
 - Market with many buyers and sellers
 - Trading identical products
 - Each buyer and seller is a price taker
 - Firms can freely enter or exit the market

Table 1

Total, Average, and Marginal Revenue for a Competitive Firm

Quantity (Q)	Price (P)	Total Revenue (TR = P x Q)	Average Revenue (AR = TR / Q)	Marginal Revenue (MR = $\Delta TR / \Delta Q$)
1 gallon	\$6	\$6	\$6	-
2	6	12	6	6
3	6	18	6	6
4	6	24	6	6
5	6	30	6	6
6	6	36	6	6
7	6	42	6	6
8	6	48	6	6

Table 2

Profit Maximization: A Numerical Example

Quantity (Q)	Total Revenue (TR)	Total Cost (TC)	Profit (TR – TC)	Marginal Revenue (MR = $\Delta TR / \Delta Q$)	Marginal Cost (MC = $\Delta TC / \Delta Q$)	Change in Profit (MR – MC)
0 gallons	\$0	\$3	-\$3	-	-	-
1	6	5	1	\$6	\$2	\$4
2	12	8	4	6	3	3
3	18	12	6	6	4	2
4	24	17	7	6	5	1
5	30	23	7	6	6	0
6	36	30	6	6	7	-1
7	42	38	4	6	8	-2
8	48	47	1	6	9	-3



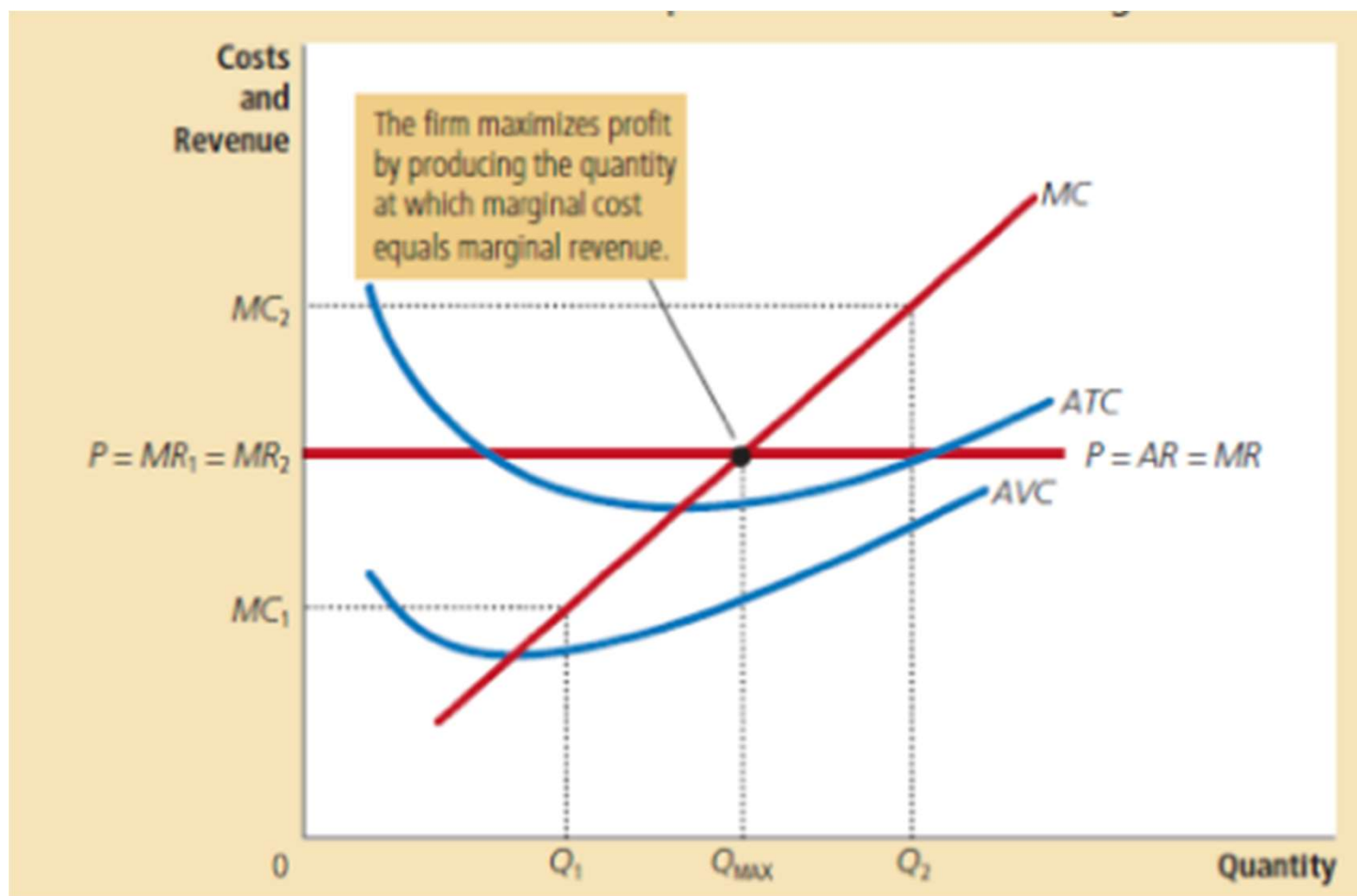
Profit Maximization

- Maximize profit
 - Produce quantity where total revenue minus total cost is greatest
 - Compare marginal revenue with marginal cost
 - If $MR > MC$: increase production
 - If $MR < MC$: decrease production
 - Maximize profit where $MR = MC$



Profit Maximization

- Rules for profit maximization:
 - If $MR > MC$, firm should increase output
 - If $MC > MR$, firm should decrease output
 - If $MR = MC$, profit-maximizing level of output

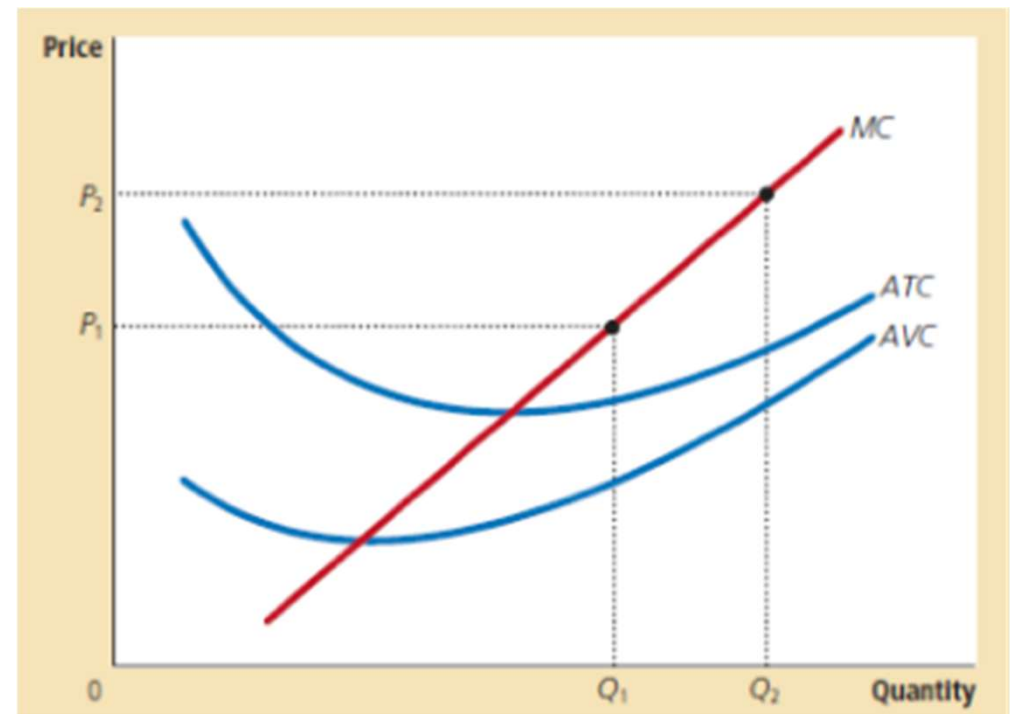




Profit Maximization

- Marginal-cost curve

- Determines the quantity of the good the firm is willing to supply at any price
- Is it supply curve?





When firm quit the market?

- Shutdown (temporary)
 - Short-run decision not to produce anything
 - During a specific period of time
 - Because of current market conditions
 - Firm still has to pay fixed costs
- Exit (permanent)
 - Long-run decision to leave the market
 - Firm doesn't have to pay any costs

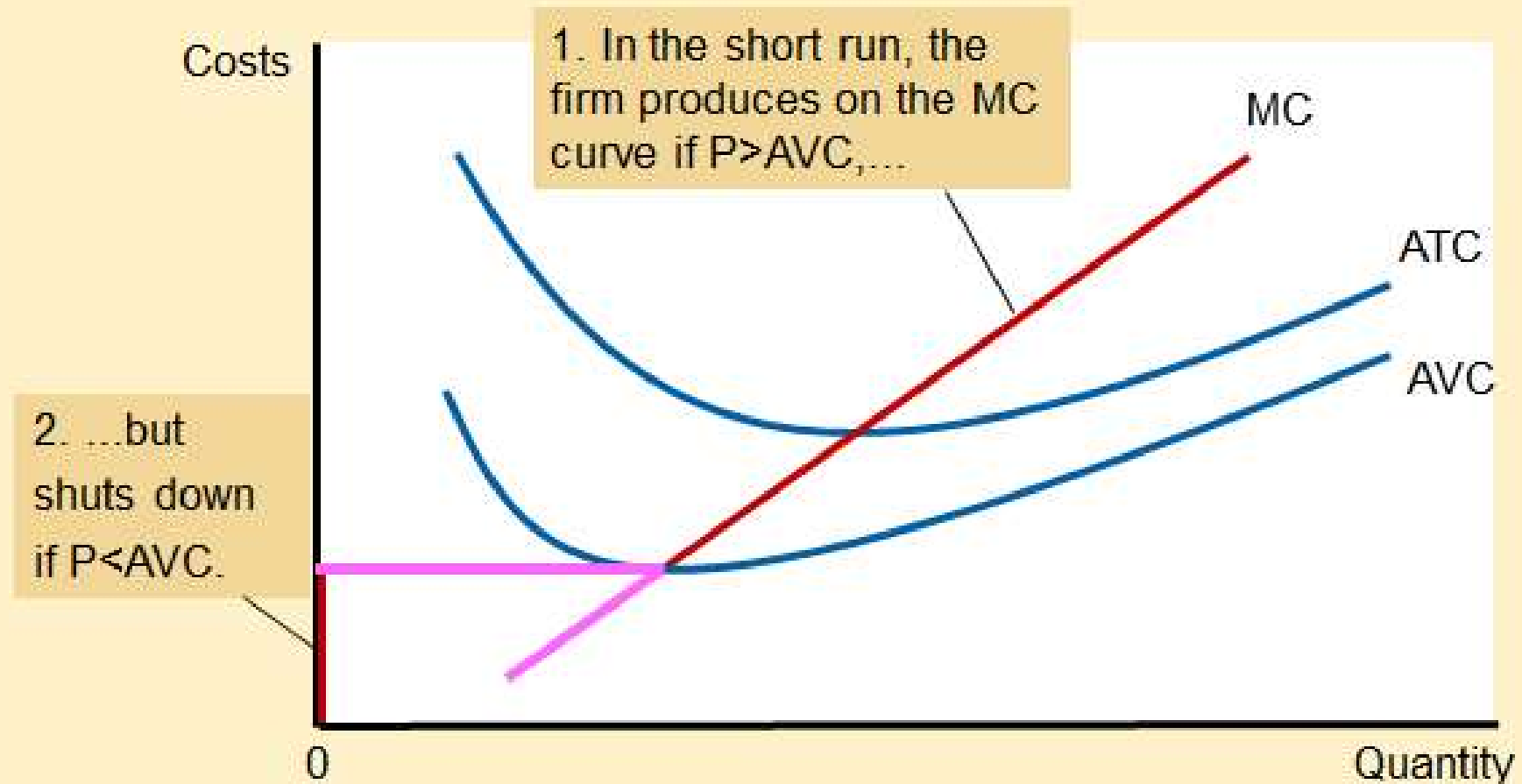


When firm quit the market?

- The firm's short-run decision to shut down
 - TR = total revenue
 - VC = variable costs
- Firm's decision:
 - Shut down if $TR < VC$ ($P < AVC$)
- Competitive firm's short-run supply curve
 - The portion of its marginal-cost curve
 - That lies above average variable cost

Figure 3

The Competitive Firm's Short-Run Supply Curve



In the short run, the competitive firm's supply curve is its marginal-cost curve (MC) above average variable cost (AVC). If the price falls below average variable cost, the firm is better off shutting down temporarily.

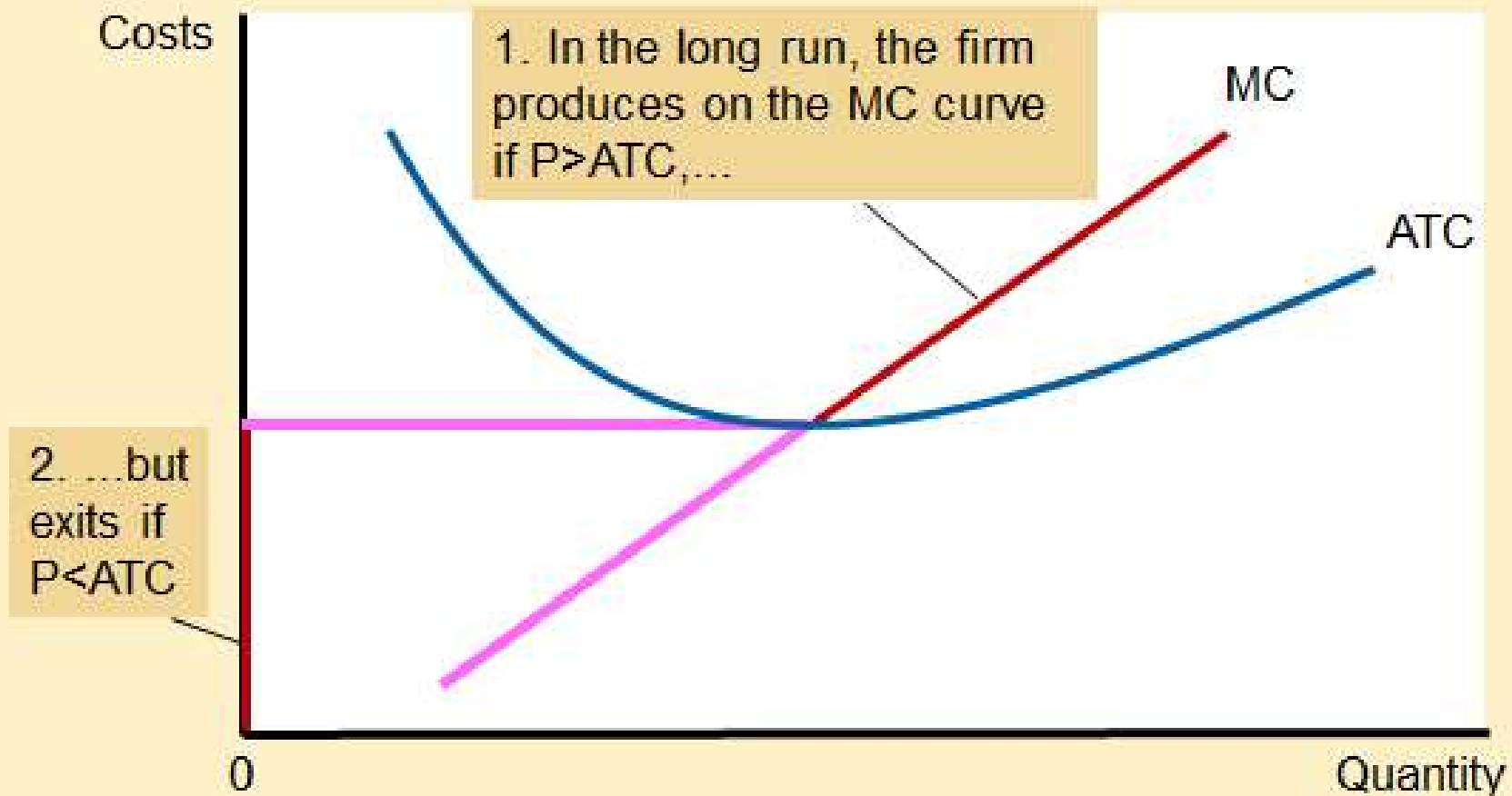


Profit Maximization, Part 7

- Firm's long-run decision
 - Exit the market if
 - Total revenue $<$ total costs; $TR < TC$
 - Same as: $P < ATC$
 - Enter the market if
 - Total revenue $>$ total costs; $TR > TC$
 - Same as: $P > ATC$
- Competitive firm's long-run supply curve
 - The portion of its marginal-cost curve that lies above average total cost

Figure 4

The Competitive Firm's Long-Run Supply Curve



In the long run, the competitive firm's supply curve is its marginal-cost curve (MC) above average total cost (ATC). If the price falls below average total cost, the firm is better off exiting the market.

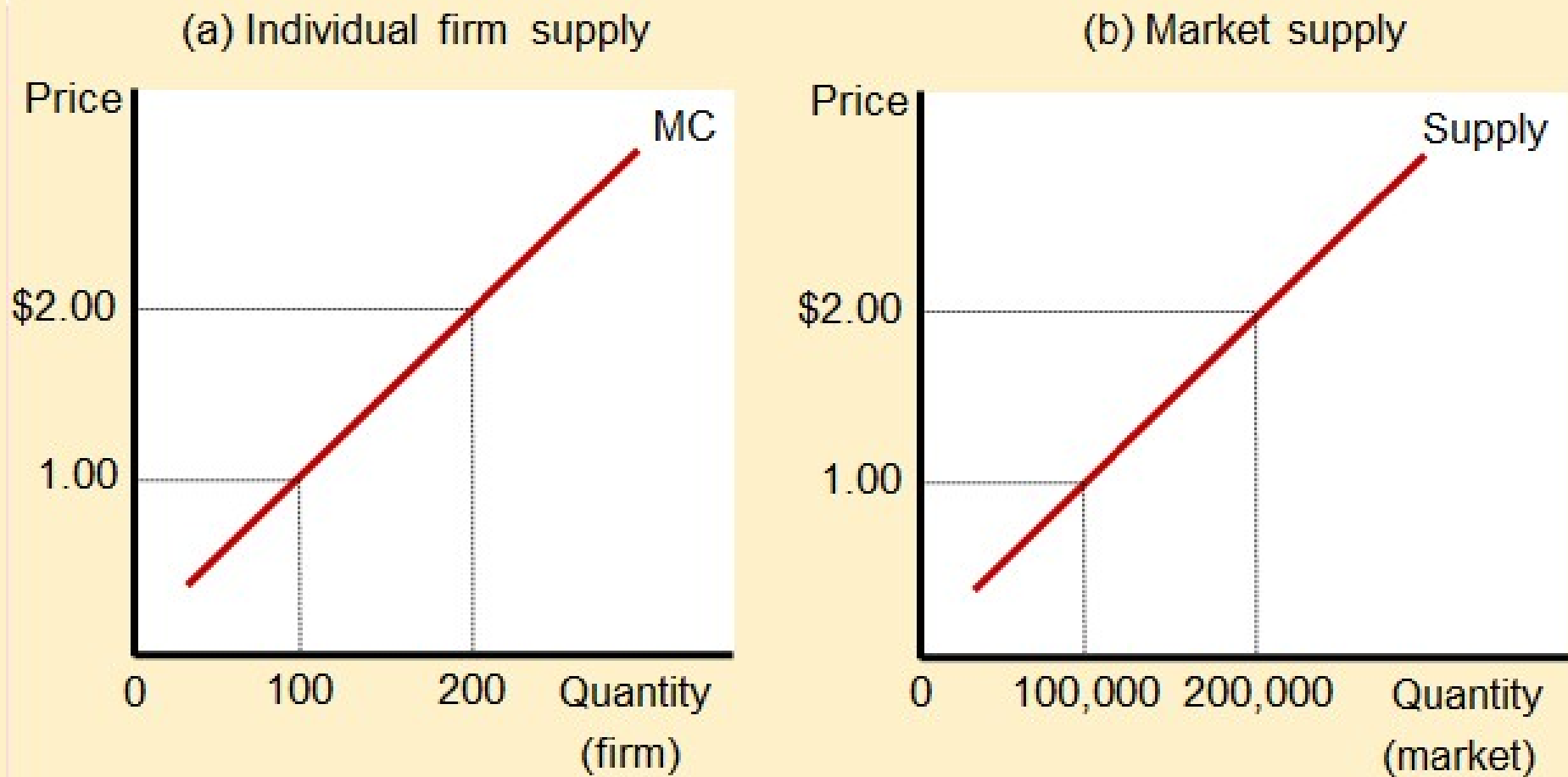


Supply Curve

- Short run: market supply with a fixed number of firms
 - Short run: number of firms is fixed
 - Each firm supplies quantity where $P = MC$
 - For $P > AVC$: supply curve is MC curve
 - Market supply
 - Add up quantity supplied by each firm

Figure 6

Short-Run Market Supply



In the short run, the number of firms in the market is fixed. As a result, the market supply curve, shown in panel (b), reflects the individual firms' marginal-cost curves, shown in panel (a). Here, in a market of 1,000 firms, the quantity of output supplied to the market is 1,000 times the quantity supplied by each firm.



Supply Curve

- Long run
 - Firms can enter and exit the market
 - If $P > ATC$, firms make positive profit
 - New firms enter the market
 - If $P < ATC$, firms make negative profit
 - Firms exit the market

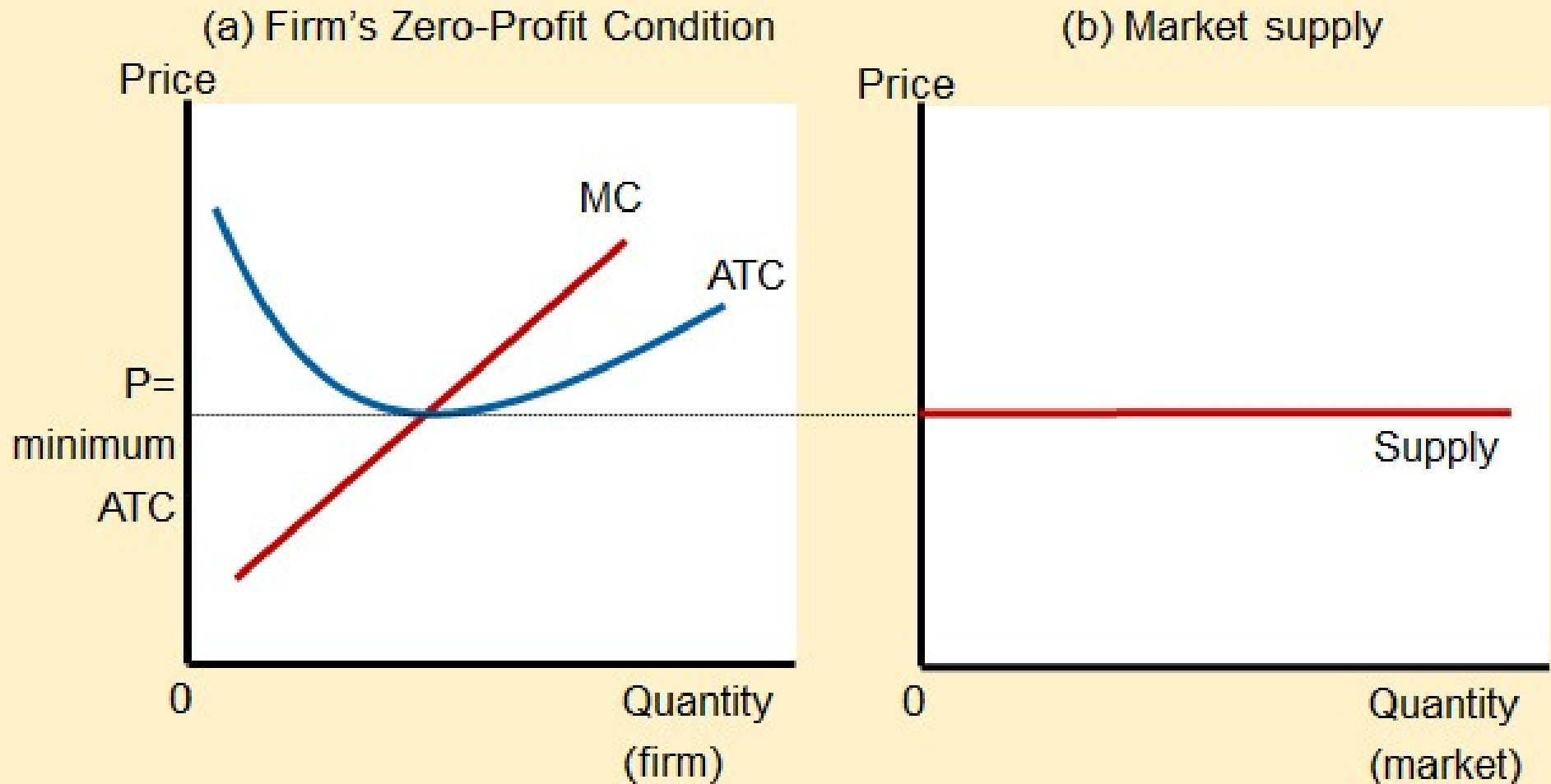


Supply Curve

- Long run
 - Process of entry and exit ends when
 - Firms still in market make zero economic profit ($P = ATC$)
 - Because $MC = ATC$: Efficient scale
 - Long run supply curve is perfectly elastic
 - Horizontal at minimum ATC

Figure 7

Long-Run Market Supply



In the long run, firms will enter or exit the market until profit is driven to zero. As a result, price equals the minimum of average total cost, as shown in panel (a). The number of firms adjusts to ensure that all demand is satisfied at this price. The long-run market supply curve is horizontal at this price, as shown in panel (b).



Supply Curve

- Why do competitive firms stay in business if they make zero profit?
 - Profit = total revenue – total cost
 - Total cost includes all opportunity costs
 - Zero-profit equilibrium
 - Economic profit is zero
 - Accounting profit is positive



"We're a nonprofit organization - we don't intend to be, but we are!"



Supply Curve

- Market in long run equilibrium
 - $P = \text{minimum ATC}$
 - Zero economic profit
- Increase in demand
 - Demand curve shifts outward
 - Short run
 - Higher quantity
 - Higher price: $P > \text{ATC}$ – positive economic profit

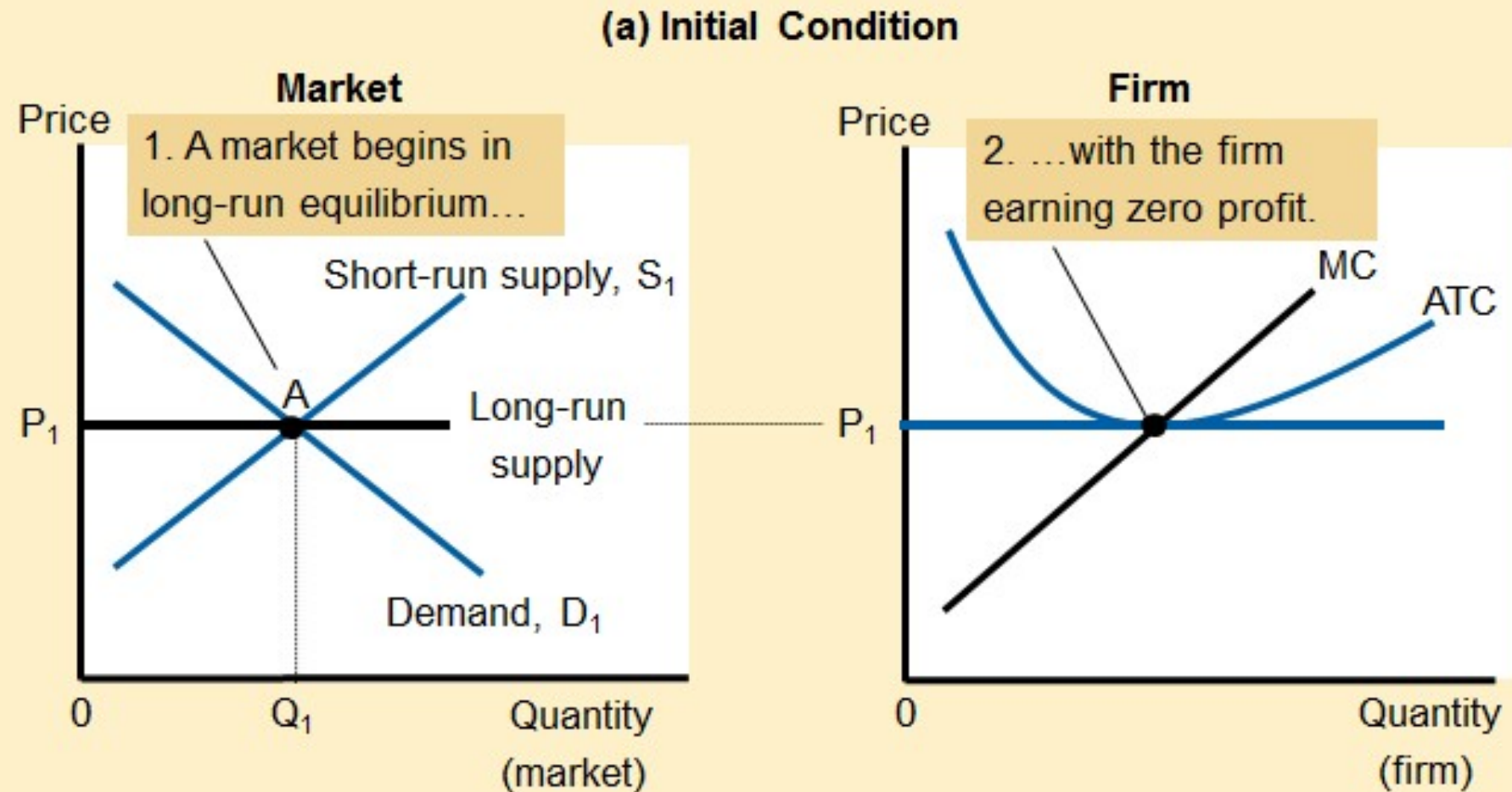


Supply Curve

- Positive economic profit in short run
 - Long run – firms enter the market
 - Short run supply curve – shifts right
 - Price – decreases back to minimum ATC
 - Quantity – increases
 - Because there are more firms in the market
 - Efficient scale

Figure 8, Part 1

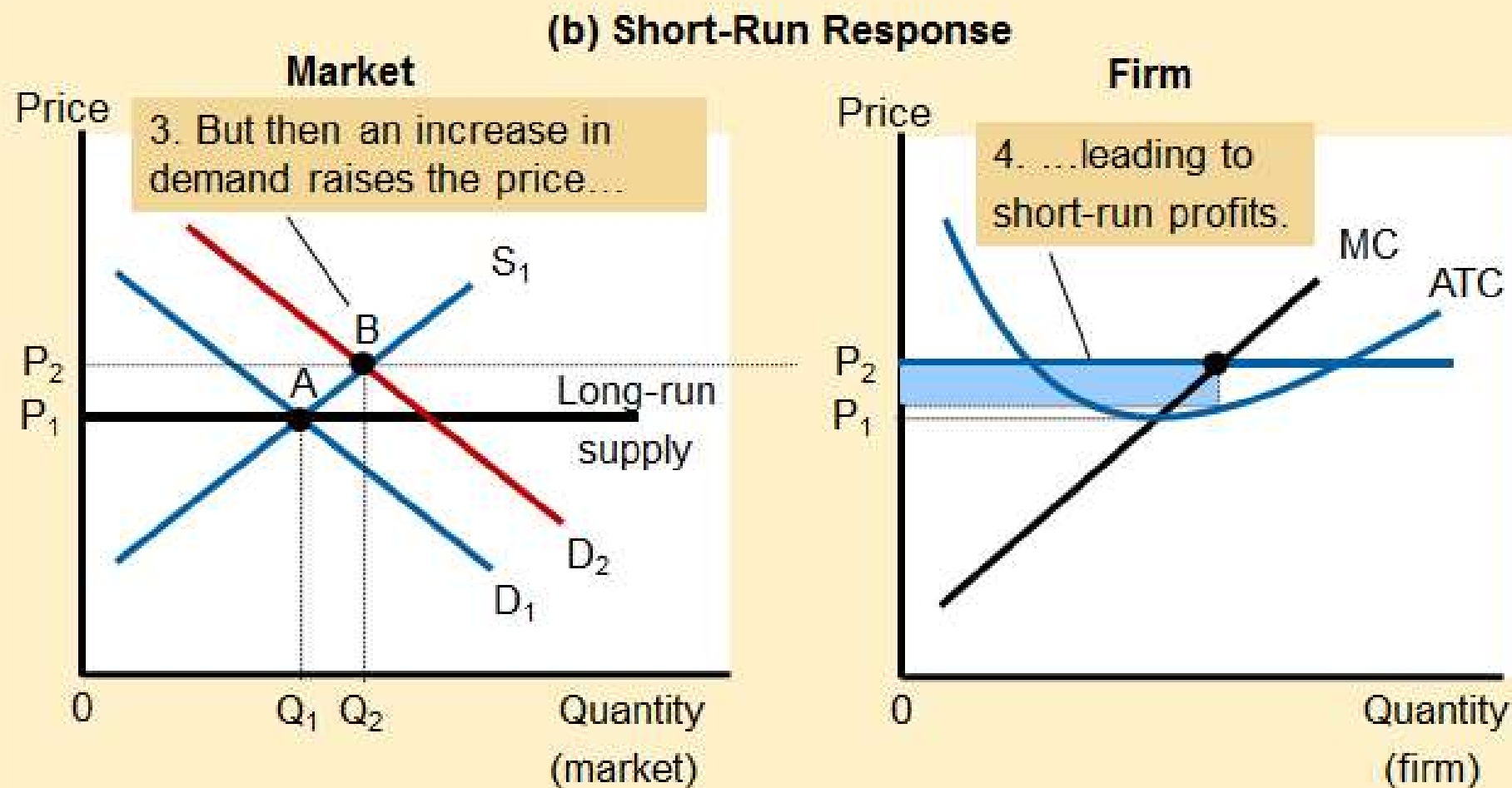
An Increase in Demand in the Short Run and Long Run (a)



The market starts in a long-run equilibrium, shown as point A in panel (a). In this equilibrium, each firm makes zero profit, and the price equals the minimum average total cost.

Figure 8, Part 2

An Increase in Demand in the Short Run and Long Run (b)

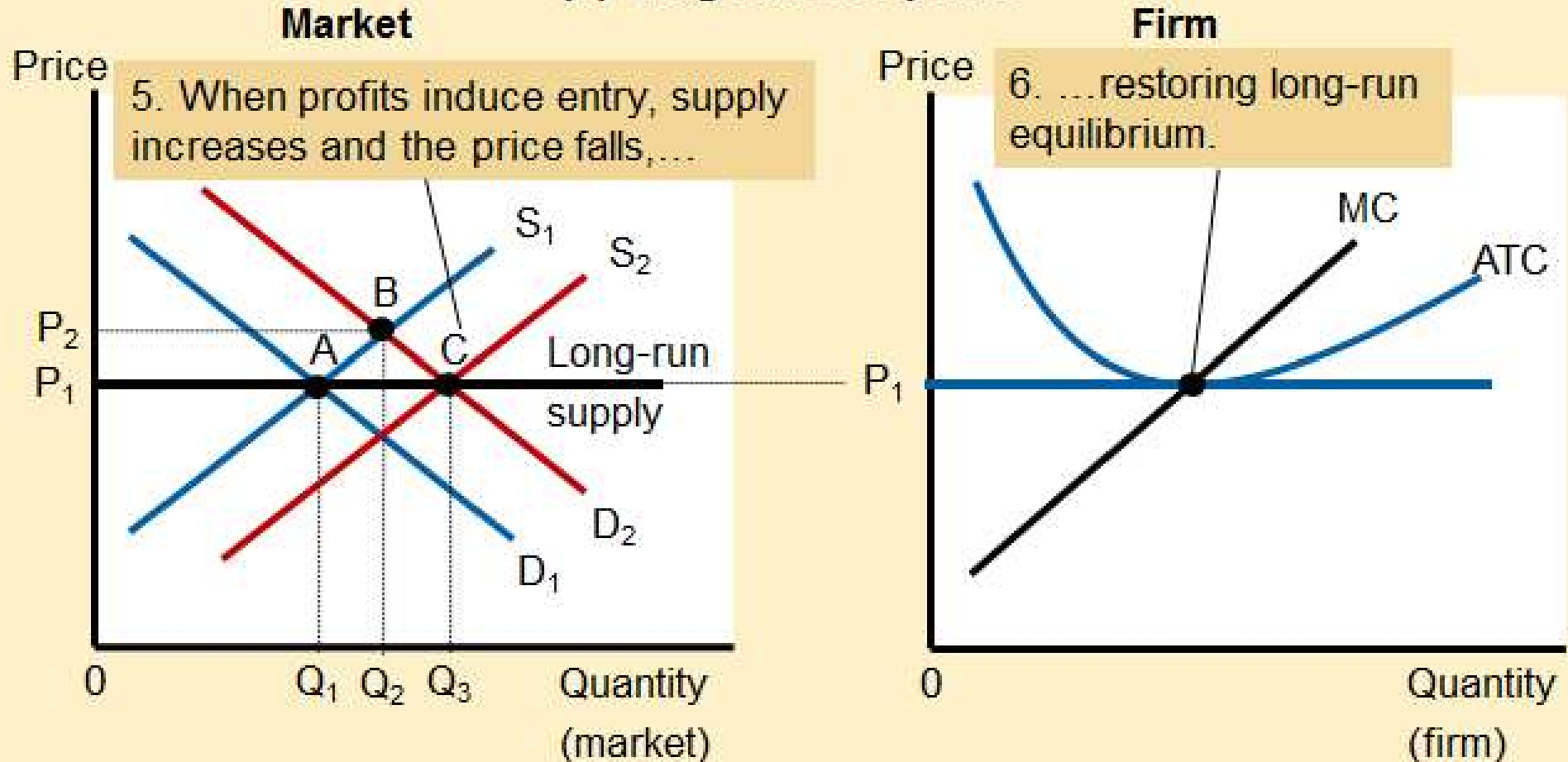


Panel (b) shows what happens in the short run when demand rises from D_1 to D_2 . The equilibrium goes from point A to point B, price rises from P_1 to P_2 , and the quantity sold in the market rises from Q_1 to Q_2 . Because price now exceeds average total cost, each firm now makes a profit, which over time encourages new firms to enter the market.

Figure 8, Part 3

An Increase in Demand in the Short Run and Long Run (c)

(c) Long-Run Response



This entry shifts the short-run supply curve to the right from S_1 to S_2 , as shown in panel (c). In the new long-run equilibrium, point C, price has returned to P_1 but the quantity sold has increased to Q_3 . Profits are again zero, and price is back to the minimum of average total cost, but the market has more firms to satisfy the greater demand.